数值分析方法笔记

数值分析

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数值分析

摘 要

1 绪论

数值分析方法，侧重的是方法，更偏向于计算

1.1 数值分析在做什么

1. 近似解 vs 精确解
2. 迭代算法
3. 收敛性
4. 误差分析

1.2 重大科学问题

1. 人工智能低能耗问题
2. 非线性与尺度的关系

1.3 数值分析

1. Numerical approaches for
   (a) linear equations
   (b) nonlinear equations
   (c) differentiation and integration
   (d) partial differential equations

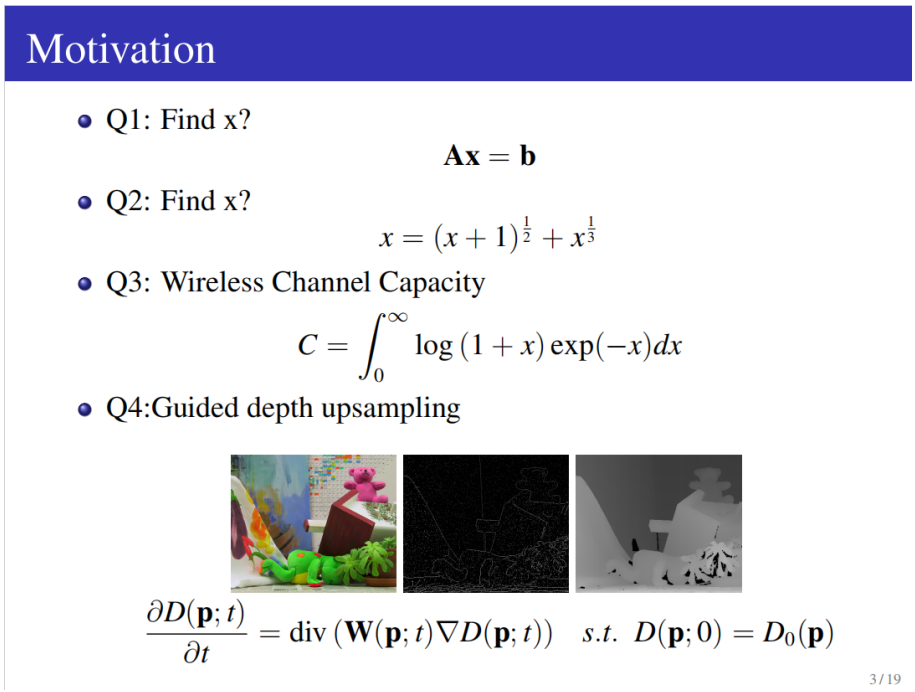

图 1: motivation

2. Error analysis

2 Computer Arithmetic

Basics: 1 Byte(字节) = 8 Bits(比特)

2.1 数据类型

1. short real format (4 Bytes)
2. long real format (8 Bytes)
   (a) Default data type in MATLAB, double in C language

- Long real format
  - Default data type in MATLAB, 'double' in C
  - Base: 2

1 bit	11 bits	52 bits
s	c	f

$$(-1)^s 2^{c-1023}(1+f)$$

  - Ex: 0 10000000011 101110 . . . 0
  - Maximum / $2^{1023} \cdot (2-2^{-52}) \approx 0.17977 \times 10^{309}$
  - Minimum / $2^{-1022} \cdot (1+0) \approx 0.22251 \times 10^{-307}$
  - Overflow / underflow
  - http://en.wikipedia.org/wiki/IEEE_floating_point

图 2: Long-real-format

(b) 会发现，能表示的最大位数并不是想象中的 2^{1024} 是因为占用了最高位和最低位 Inf and NaN

(c) 在有效位数截断时，有两种方式，一种直接丢弃 (way of chopping)，另一种则是在对应位加上 5 再舍去 (way of Rooundoff)

- Base: 10
- k-digit decimal machine number:

$$\pm 0.d_1 d_2 \ldots d_k \times 10^n, \quad 1 \le d_1 \le 9, \quad 0 \le d_i \le 9$$

- Any positive number within the numerical range can be writte

$$y = 0.d_1 d_2 \ldots d_k d_{k+1} d_{k+2} \ldots \times 10^n$$

- Two ways to represent y with k digits:
  1. Chopping:
  $$fl(y) = 0.d_1 d_2 \ldots d_k \times 10^n$$
  2. Roundoff: Add $5 \times 10^{n-(k+1)}$ and chop:
  $$fl(y) = 0.\delta_1 \delta_2 \ldots \delta_k \times 10^n$$
- Roundoff error

图 3: ERROR-ANALYSIS

2.2 ERRORS and significant digits

2.2.1 Definations

定义 2.1. *If p^* is an approximation to p, the* ***absolute error*** *is $|p-p^*|$, and the* ***relative error*** *is $\frac{|p-p^*|}{|p|}$, provided that $p \neq 0$.*

Significant digits:

The number p^* is said to approximate p to t significant digits if t is the largest nonnegative integer for which

$$\frac{|p-p^*|}{|p|} \leq 5 \times 10^{-t}$$

$$\begin{aligned}
\left|\frac{y - fl(y)}{y}\right| &= \left|\frac{0.d_1d_2\ldots d_kd_{k+1}\ldots \times 10^n - 0.d_1d_2\ldots d_k \times 10^n}{0.d_1d_2\ldots \times 10^n}\right| \\
&= \left|\frac{0.d_{k+1}d_{k+2}\ldots \times 10^{n-k}}{0.d_1d_2\ldots \times 10^n}\right| = \left|\frac{0.d_{k+1}d_{k+2}\ldots}{0.d_1d_2\ldots}\right| \times 10^{-k} \\
&\leq \frac{1}{0.1} \times 10^{-k} = 10^{-k+1}
\end{aligned}$$

图 4: ErrorAnalysis.png

疑问，这里为什么是 5 解答：这里的 5 是 define，作为误差判定的标准

- Floating Point Operations
  - Machine addition: $x \oplus y = fl(fl(x) + fl(y))$
  - Subtraction: $x \ominus y = fl(fl(x) - fl(y))$
  - Multiplication: $x \otimes y = fl(fl(x) \times fl(y))$
  - division: $x \oslash y = fl(fl(x) \div fl(y))$
- Cancelation
  - Common problem: Subtraction of nearly equal numbers:

$$fl(x) = 0.d_1d_2\ldots d_p\alpha_{p+1}\alpha_{p+2}\ldots\alpha_k \times 10^n$$

$$fl(y) = 0.d_1d_2\ldots d_p\beta_{p+1}\beta_{p+2}\ldots\beta_k \times 10^n$$

gives fewer digits for significance:

$$fl(fl(x) - fl(y)) = 0.\sigma_{p+1}\sigma_{p+2}\ldots\sigma_k \times 10^{n-p}$$

图 5: 减法消除有效位数

Given $x = 5/7$, $u = 0.714251$, $v = 98765.9$, and $w = 0.111111 \times 10^{-4}$. Determine the five-digit chopping values of

Operation	Result	Actual value	Absolute error	Relative error
$x \ominus u$	0.30000×10^{-4}	0.34714×10^{-4}	0.471×10^{-5}	0.136
$(x \ominus u) \oplus w$	0.27000×10^{1}	0.31242×10^{1}	0.424	0.136
$(x \ominus u) \otimes v$	0.29629×10^{1}	0.34285×10^{1}	0.465	0.136
$u \oplus v$	0.98765×10^{5}	0.98766×10^{5}	0.161×10^{1}	0.163×10^{-4}

图 6: 不同运算误差差别很大

考虑一个 equation $x^2 + 62.10x + 1 = 0$, 如果我们采用四舍五入算法保留一定的有效位数（四位），最后的误差达到了 24%，非常高; **但是但是，如果将无理数放在分母上，计算误差就会降低到** 6.2×10^{-4}

2.3 Algorithms(收敛)

a Algorithms is a set if operation to solve a problem or approximate a solution to ze problem (迭代的收敛性)

$$|\alpha_n - \alpha| \leq K|\beta_n|$$

收敛速度迭代

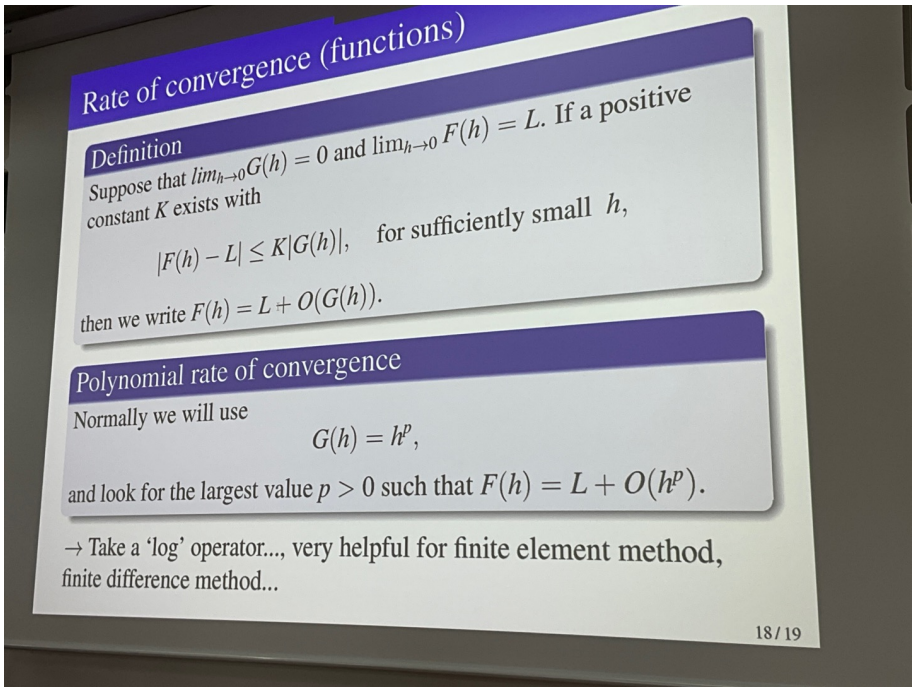

图 7: 收敛速度 2.jpg

3 Solutions of Equations in One Variable(单变量求根)

参考文章：知乎

定理 3.1. *Rolle's Theorem(罗尔中值定理)*

定理 3.2. *Mean Value Theorem*（又叫拉格朗日中值定理）

定理 3.3. *Intermedi Value Theorem*（介值定理）

3.1 The Way of Root Finding Problem

Forward problem

Inverse problem

1. 二分法 (The Bisection Method)(Because of IVT)
2. 不动点法
3. 牛顿法

3.2 The Bisection Method

3.2.1 Method

在区间 (a,b) 中有零点，求出区间中点 $\frac{a+b}{2}$ 的函数值，记为 $f(x_1)$，那么当 $f(x_1)=0$，证明找到零点了，迭代结束，不是的话，假设 $f(a)f(x_1)<0$，则在 (a,x_1) 中迭代求解✦，当然也可能 $f(b)f(x_1)<0$，那么就在 (x_1,b) 中迭代求解。

3.2.2 Convergence

If the root lies at r, and after n iterations the interval is $[a_n, b_n]$, the length of the interval after each iteration is halved, i.e.,

$$|b_n - a_n| = \frac{|b_0 - a_0|}{2^n}.$$

Let ε_n denote the error at the n-th iteration, which is the distance between the midpoint and the actual root r. After n iterations, the error can be bounded by:

$$\varepsilon_n = |c_n - r| \leq \frac{|b_0 - a_0|}{2^n}.$$

Therefore, after n iterations, the maximum possible error decreases exponentially by a factor of $\frac{1}{2}$ in each step, meaning the method converges **linearly**, with a convergence rate 1/2.

$$x_n = x + O(\frac{1}{2^n})$$

,the rate of Convergence is(收敛速度) 为

$$O(\frac{1}{2^n})$$

3.3 The Fixed-Point Problem(不动点法)

For function $f(p) = 0$,convert it to $f(p) = g(p) - p$,then the problem convert to The Fixed-Point Problem

3.4 Well-Posed(适定性)

适定性存在、唯一且连续

3.4.1 Conception（概念）

1.2 不动点迭代

首先要明白什么是不动点，不动点的定义为：

若 $f(r) = r$ ，则称 r 为 $f(r)$ 的一个不动点。

那问题来了，明明是函数求零点为什么要扯到不动点上去？

上面可以看出，二分法迭代速度慢，而不动点的优势恰恰是快速迭代求解。

对于不动点有下面的描述成立：

$f \in [a,b]$ 且在 $[a,b]$ 上为自身映射[+]（f 在 $[a,b]$ 上的值域包含于 $[a,b]$ ，同济高数第一章介绍过），则其在 $[a,b]$ 中存在不动点。特别地，当 f 可微且 $\exists 0 < k < 1, \forall x \in [a,b], |f^{'}(x)| \leq k$ 时，不动点的存在性是唯一的。

对于不动点迭代：选择初值 x_0 ，构造数列 $x_n = f(x_{n-1})$ ，则 $\lim_{n\to\infty} x_n = x$ 为不动点。

不动点迭代[+]的几何意义相当于从 $(x_0, 0)$ 作竖直直线交 f 于(x_0, x_1)，再由(x_0, x_1)作水平直线交 $y = x$ 于 (x_1, x_1) ，再由(x_1, x_1)作竖直直线交 f 于 (x_1, x_2) ,不断重复，直至收敛。

强调，只有是稳定不动点（对于 k 的要求在（0,1）范围内）的时候，不动点迭代才会收敛（压缩映射原理）

3.4.2 不动点收敛速度

In the condition of $|g'(x)| \leq k < 1$, for $\forall x \in [a, b]$,we have

$$|p_n - p| = |g(p_{n-1}) - p| = |g'(\xi)||p_{n-1} - p|$$

$$\leq k|p_{n-1} - p|$$

$$\leq k^2|p_{n-2} - p|$$

$$\leq k^n|p_1 - p|$$

and it's obviously that the p_n is astringent（收敛的），and this Theorem is called Contraction Mapping (压缩映射原理)

思考：为什么逆问题这么困难——> 因为逆问题的时间复杂度要比正问题高得多（举例：求矩阵的逆）

3.5 Newton's Method

3.5.1 Conceptioin

考虑 Taylor 公式，在根附近展开，我们有

$$0 = f(p_0) + f'(p_0)(p - p_0)$$

由此得到

$$p = p_0 - \frac{f(p_0)}{f'(p_0)}$$

迭代公式转化为

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$$

记

$$g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$$

则不难发现 Netwon's Method 其实也是一种特殊的不动点法

3.5.2 收敛性判断

对于一般的不动点法来说，要求 $|g'(x)| \leq k < 1$，那我们来看 Netwon's Method, 有

$$g'(x) = ... = \frac{f(x)f''(x)}{(f'(x))^2}$$

与此同时，由于 $f(x_0)=0$, 那么我们就有 $g'(x_0)=0$, 那么

$$\exists\delta>0,\forall x\in[x_0-\delta,x_0+\delta],\text{我们总有}g'(x)=k<1$$

这时就会发现在零点附近，Netwon's Method 一定是收敛的不动点而不是发散的，而且收敛速度也会很快

3.5.3 Adjoint Method（伴随方法 or BT（即神经网络中的反向传播梯度算法））

Netwon's Method 方法收敛速度很快，但是他也给计算机计算带来了问题——导数的计算也是存在一定的困难，那么这时，引入 BT 算法就会在一定程度上解决这个问题（BT 算法也是很不容易）**另：Hessian(求解二阶梯度) 方法**

3.6 Order of Convergence

Order of Convergence

Suppose $\{p_n\}_{n=0}^{\infty}$ is a sequence that converges to p, with $p_n\neq p$ for all n. If positive constants λ and α exist with

$$\lim_{n\to\infty}\frac{|p_{n+1}-p|}{|p_n-p|^{\alpha}}=\lambda$$

then $\{p_n\}_{n=0}^{\infty}$ converges to p of order α, with asymptotic error constant λ.

- If $\alpha=1$, the sequence is linearly convergent.
- If $\alpha=2$, the sequence is quadratically convergent.

图 8: Order of Convergence

3.6.1 Error Analysis for Fix-point Methods

Proof

- Since g' exists on (a,b), applying the MVT, we have

$$p_{n+1}-p=g(p_n)-g(p)=g'(\xi_n)(p_n-p)$$

- Since $\{p_n\}_{n=0}^{\infty}$ converges to p, $\{\xi_n\}_{n=0}^{\infty}$ also converges to p.
- Thus,

$$\lim_{n\to\infty}\frac{p_{n+1}-p}{p_n-p}=\lim_{n\to\infty}g'(\xi_n)=g'(p)$$

Hence, if $g'(p)\neq 0$, fixed-point iteration exhibits linear convergence with asymptotic error constant $|g'(p)|$.

图 9: Error Analysis for Fix-point Methods

Proof (2/2)

- When $x = p_n$, we have

$$p_{n+1} = g(p_n) = p + \frac{g''(\xi)}{2}(p_n - p)^2$$

- Thus,

$$\lim_{n\to\infty} \frac{|p_{n+1} - p|}{|p_n - p|^2} = \frac{|g''(p)|}{2} < \frac{M}{2}$$

图 10: Netwon's Methods

3.6.2 Error Analysis for Netwon's Methods

4 Fixed Points for Functions of Several Variables

4.1 Vector Norm(向量范数)

Tips: 此处所指向量不加说明默认为列向量

4.1.1 Defination

范数的一些基本性质（向量空间中范数的定义以及在线性空间中体现出来的性质）

Definition: Vector Norm

A vector norm on $\mathbb{R}^n$ is a function, $||\cdot||$, from $\mathbb{R}^n$ into $\mathbb{R}$ with the following properties:

1. $||\mathbf{x}|| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$
2. $||\mathbf{x}|| = 0$ iff $\mathbf{x} = 0$
3. $||\alpha\mathbf{x}|| = |\alpha|||\mathbf{x}||$ for all $\alpha \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$
4. $||\mathbf{x} + \mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}||$ for all $||\mathbf{x}||, ||\mathbf{y}|| \in \mathbb{R}^n$.

图 11: DefVectorNorm

范数不同的定义

Definition: L_1, L_2, and L_∞ Norms

The norms for the vector $\mathbf{x} = (x_1, x_2, \cdots, x_n)^t$ are defined by:

- L_1 Norm
$$||\mathbf{x}||_1 = \sum_{i=1}^{n} |x_i|$$
- L_2 Norm
$$||\mathbf{x}||_2 = \left\{ \sum_{i=1}^{n} x_i^2 \right\}^{1/2}$$
- L_∞ Norm
$$||\mathbf{x}||_\infty = \max_{1 \le i \le n} |x_i|$$

Note that each of these norms reduces to the absolute value in the case $n = 1$.

图 12: Vector Norm

4.2 A Limit of a Sequence of Vectors in $\mathbb{R}^n$

定理 4.1. *For each* $\mathbf{x} \in \mathbb{R}^n$, *wehave* $\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\|x\|_\infty$

This Theorem is very easy, we will less introduce the proof of it,here is a picture to show the answer:

Proof

Let x_j be a coordinate of $\mathbf{x}$ such that $||\mathbf{x}||_\infty = \max_{1 \le i \le n} |x_i| = |x_j|$. Then
$$||\mathbf{x}||_\infty^2 = |x_j|^2 \le \sum_{i=1}^{n} x_i^2 = ||\mathbf{x}||_2^2$$
and
$$||\mathbf{x}||_\infty \le ||\mathbf{x}||_2$$
so
$$||\mathbf{x}||_2^2 = \sum_{i=1}^{n} x_i^2 \le \sum_{i=1}^{n} x_j^2 = n x_j^2 = n||\mathbf{x}||_\infty^2$$
and $||\mathbf{x}||_2 \le \sqrt{n}||\mathbf{x}||_\infty$.

图 13: proof

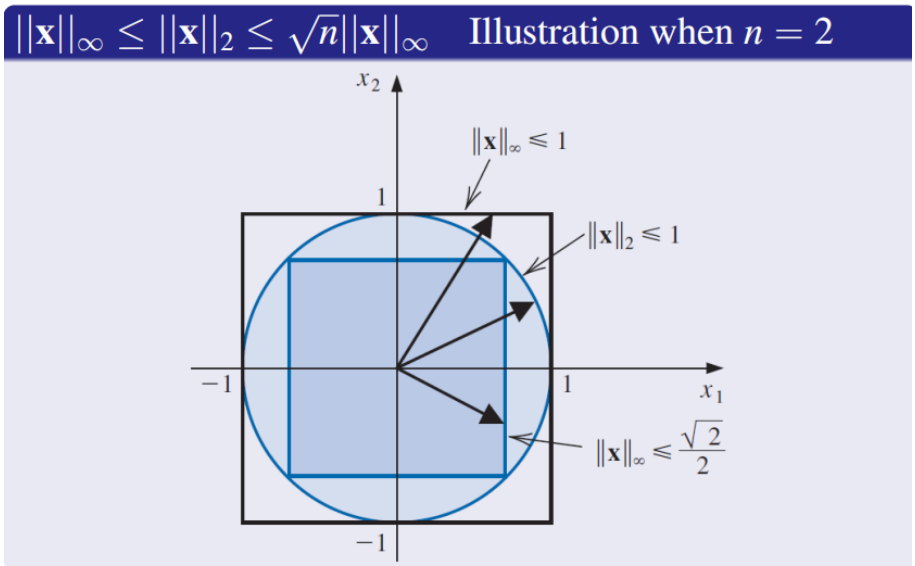

图 14: picsstures to proof

定理 4.2. *For each*

4.3 Fixed Points for Functions of Several Variables

A system of nonlinear equations has the form:

$$f_1(x_1, x_2, x_3, ..., x_n) = 0,$$

$$f_2(x_1, x_2, x_3, ..., x_n) = 0,$$

$$f_3(x_1, x_2, x_3, ..., x_n) = 0,$$

...

$$f_n(x_1, x_2, x_3, ..., x_n) = 0,$$

which can be represented by

$$\mathbf{F}(\mathbf{x}) = \mathbf{0}$$

The functions f1, f2, . . . , fn are called the coordinate functions(坐标函数) of F.

但是要注意，n 个方程要求是相互独立的（岂不是废话 bushi）

4.4 Fix-Point Method

A function $\mathbf{G}$ from $\mathbf{D} \in \mathbb{R}^n$ into R n has a fixed point at p ∈ D if $\mathbf{G(p)} = \mathbf{p}$.

Fixed Points in $\mathbb{R}^n$

A function $\mathbf{G}$ from $\mathbf{D} \in \mathbb{R}^n$ into $\mathbb{R}^n$ has a fixed point at $\mathbf{p} \in D$ if $\mathbf{G}(\mathbf{p}) = \mathbf{p}$.

Theorem (Existence of Fixed Points)

Let $\mathbf{D} = \{(x_1, x_2, \ldots, x_n)^t | a_i \leq x_i \leq b_i, \text{for each } i = 1, 2, \ldots, n\}$ for some collection of constants $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$. Suppose $\mathbf{G}$ is a continuous function from $\mathbf{D} \in \mathbb{R}^n$ into $\mathbb{R}^n$ with the property that $\mathbf{G}(\mathbf{x}) \in \mathbf{D}$ whenever $\mathbf{x} \in \mathbf{D}$. Then $\mathbf{G}$ has a fixed point in $\mathbf{D}$.

图 15: fixpoint

Theorem (Fixed Point Theorem)

Let $\mathbf{D} = \{(x_1, x_2, \ldots, x_n)^t | a_i \leq x_i \leq b_i, \text{for each} i = 1, 2, \ldots, n\}$ for some collection of constants $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$. Suppose $\mathbf{G}$ is a continuous function from $\mathbf{D} \in \mathbb{R}^n$ into $\mathbb{R}^n$ with the property that $\mathbf{G}(\mathbf{x}) \in \mathbf{D}$ whenever $\mathbf{x} \in \mathbf{D}$. Then $\mathbf{G}$ has a fixed point in $\mathbf{D}$.
Moreover, suppose that all the component functions of $\mathbf{G}$ have continuous partial derivatives and a constant $K < 1$ exists with

$$\left| \frac{\partial g_i(\mathbf{x})}{\partial x_j} \right| \leq \frac{K}{n}, \quad \text{whenever } \mathbf{x} \in \mathbf{D},$$

for each $j = 1, 2, \ldots, n$ and each component function g_i. Then the sequence $\{\mathbf{x}^{(k)}\}_{k=0}^{\infty}$ defined by an arbitrarily selected $\mathbf{x}^{(0)}$ in $\mathbf{D}$ and generated by

$$\mathbf{x}^{(k)} = \mathbf{G}(\mathbf{x}^{(k-1)}), \quad \text{for each } k \geq 1$$

converges to the unique fixed point $\mathbf{p} \in \mathbf{D}$ and

$$||\mathbf{x}^{(k)} - \mathbf{p}||_\infty \leq \frac{K^k}{1-K} ||\mathbf{x}^{(1)} - \mathbf{x}^{(0)}||_\infty$$

图 16: fix-point-define

原理推导 ：

$\boldsymbol{D}$ 是一个 n 元列向量，那对于从 $\boldsymbol{D}$ 到 $\boldsymbol{D}$ 映射的函数 $\boldsymbol{G}$ 而言，如果存在不动“点” $\boldsymbol{p}$，使得

$$\boldsymbol{p} = \boldsymbol{G}(\boldsymbol{p})\text{亦即}p_i = g_i(\boldsymbol{p}), \forall i = 1, 2, 3, ..., n$$

那么我们接下来讨论其收敛特性：类比一元函数的不动点收敛的充分条件，仍采用类似压缩映射 +MVT（拉格朗日中值定理）的思路，使用多元的 MVT 公式：

$$f(\boldsymbol{x}) - f(\boldsymbol{y}) = \nabla f(\boldsymbol{\zeta}) \cdot (\boldsymbol{x} - \boldsymbol{y})$$

将上式变形，并使用 g_i 来代替 f, 并使用柯西不等式可以得到：

$$\|g_i(\boldsymbol{x}) - g_i(\boldsymbol{y})\|_2 = \|\nabla f(\boldsymbol{\zeta}) \cdot (\boldsymbol{x} - \boldsymbol{y})\|_2 \leq \|\nabla f(\boldsymbol{\zeta})\|_2 \cdot \|\boldsymbol{x} - \boldsymbol{y}\|_2$$

此时我们将 $\boldsymbol{y} =$ 不动点 $\boldsymbol{p}$ 带入得到：

$$\|g_i(\boldsymbol{x}) - g_i(\boldsymbol{p})\|_2 \leq \|\nabla f(\boldsymbol{\zeta})\|_2 \cdot \|\boldsymbol{x} - \boldsymbol{p}\|_2$$

由于 $g_i(\boldsymbol{p}) = \boldsymbol{p}$ 进而有

$$\|g_i(\boldsymbol{x}) - \boldsymbol{p}\|_2 \leq \|\nabla f(\boldsymbol{\zeta})\|_2 \cdot \|\boldsymbol{x} - \boldsymbol{p}\|_2$$

即

$$\frac{\|g_i(\boldsymbol{x}) - \boldsymbol{p}\|_2}{\|\boldsymbol{x} - \boldsymbol{p}\|_2} \leq \|\nabla f(\boldsymbol{\zeta})\|_2 \triangleq K$$

我们让 K 小于 1，那么就不难推出这样的迭代是收敛的 (公式写的有点问题，意会一下，大致思路是这样)

4.5 Newton's Method

引理 4.3.

推论 4.4. *asdf*

命题 4.5. *adf*

定义 4.6. *adsf*

例 4.7. *asdfasfd*

证明. adsfasfd □

在一个线性电路中，电流的积分与电压的积分之和等于零。

asdfas[1]

Code Listing 1: Python example

```python
def hello_world():
    print("Hello, world!")
```

参考文献

[1] John Smith and Jane Doe. Deep learning for computer vision: A brief review. *Computational Intelligence and Neuroscience*, 2019:1–13, 2019.